

Summary for final exam, semester of 2025-2026

The concept of wave function

$$\Psi(x, t)$$

which satisfies

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H} \Psi$$

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}, t)$$

The probability density

$$\rho = |\Psi(x, t)|^2$$

Compare with the classical mechanics

$$\{x, p\} = 1 \implies [\hat{x}, \hat{p}] = i\hbar$$

Question: Given a wave function $\Psi(x)$

We need to

1. normalize the wave function
2. Compute $\langle \hat{x} \rangle$, $\langle \hat{p} \rangle$, $\langle \hat{x}^2 \rangle$, $\langle \hat{p}^2 \rangle$
3. Compute σ_x , σ_p , and check the uncertainty principle

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}$$

When V does not depend on time, important case: The **Stationary State**

coming from $\hat{H}\psi_n = E_n\psi_n$

which evolves with time as $\psi_n e^{-iE_n t/\hbar}$

For a general state,

$$\Psi(x, t) = \sum_n C_n \psi_n(x) e^{-iE_n t/\hbar}$$

Two very important situations when

$$1. V(x) = \begin{cases} 0 & x \in [0, a] \\ \infty & \text{otherwise} \end{cases}$$

$$2. V(x) = \frac{1}{2}m\omega^2 x^2$$

In particular for infinite square well, we need to know all the details to solve for the eigen functions and eigen values. For HO (Harmonic Oscillation, 重排者注), we have to learn how to use $\hat{a}^+$ and $\hat{a}^-$ to compute various physical quantities!

With regard to the general **formalism**, several important points are in order.

1. Hilbert space.
2. Inner product (geometrical interpretation)
3. Hermitian operator (Hermitian conjugate, properties of eigen functions and eigen values, completeness of the eigen states)
4. Commutator calculations

$$[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$$

5. Dirac notation. $|\alpha\rangle, \langle\beta|$:

$$O_{\alpha\beta} = \langle\alpha|\hat{O}|\beta\rangle \longrightarrow \text{matrix element of the operator}$$

6. Different bases in Dirac notation:

- position space: $\langle x|x'\rangle = \delta(x - x')$
- momentum space: $\langle p|p'\rangle = \delta(p - p')$

$$\langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar}$$

$$\int dx |x\rangle\langle x| = \int dp |p\rangle\langle p| = \hat{I}$$

General theory of the **angular momentum**.

$$[\hat{L}_i, \hat{L}_j] = i\hbar\epsilon_{ijk}\hat{L}_k, \quad \text{with } \vec{L} = \vec{r} \times \vec{p}$$

the spectra can be obtained from algebraic approach.

Since $[\hat{L}_z, \vec{L}^2] = 0$ (why?)

We can choose the common eigen states of $\hat{L}_z$ and $\vec{L}^2$ which are labelled by two quantum numbers, ℓ and m . $|\ell m\rangle$

$$\hat{L}_z|\ell m\rangle = m\hbar|\ell m\rangle, \quad m = -\ell, -\ell + 1, \dots, \ell$$

$$\vec{L}^2|\ell m\rangle = \ell(\ell+1)\hbar^2|\ell m\rangle, \quad \ell = 0, \frac{1}{2}, 1, 2, \dots$$

The magnetic number m can be raised or lowered by the operator $\hat{L}_+$ or $\hat{L}_-$.

The most important case is the spin $\frac{1}{2}$ particle whose spin operator can be written in terms of Pauli matrices.

$$\vec{S} = \frac{\hbar}{2}\vec{\sigma}$$

and

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$$\sigma_y = \begin{pmatrix} 0 & i \\ -i & 0 \end{pmatrix}$$

$$\sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

A general spin state of the particle is described by

$$\chi = \begin{pmatrix} a \\ b \end{pmatrix}$$

Question: Given a general state, what is the probability of getting $\pm\hbar/2$ when measuring $\hat{S}_i$?

Adding the spin of two spin $\frac{1}{2}$ particles, the total spin can be --

- $S = 0 \rightarrow$ Singlet $\rightarrow$ Asymmetric
- $S = 1 \rightarrow$ Triplet $\rightarrow$ Symmetric

For **identical fermions**, the wave function of the system must be asymmetric with respect to the exchange of any two particles.

Which for identical bosons, the wave function must be symmetric.

Question: Give the single particle spectrum, obtain the ground state (including the wave function and the energy) and the first excited state of the system composed of two identical particles?

In general, the exact eigen function of $\hat{H}$ is difficult to obtain when V is complicated or time-dependent! This is the situation where the perturbation theory sets in!

$$\text{Perturbation Theory} \begin{cases} \text{time-independent} \\ \text{Non-degenerate spectrum (a)} \\ \text{Degenerate spectrum (b)} \\ \text{time-dependent (c)} \end{cases}$$

For (a): $H = H_0 + H'$, $H_0\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}$

Non-degenerate:

$$E_n^{(1)} = H'_{nn}$$

$$E_n^{(2)} = \sum_{m \neq n} \frac{|H'_{mn}|^2}{\Delta E_{nm}}$$

Degenerate: Find the eigen values of H' in the sub space spanned by the degenerate states corresponding to the same $E_n^{(0)}$.

These eigen values are the first order correction to the energy $E_n^{(1)}$.

Question: Given a HO, adding a perturbation H' which can be either proportional to $\hat{x}$ or $\hat{x}^2$, compute $E_n^{(1)}$, $E_n^{(2)}$?